

Subnational Immunisation Coverage & Equity — Ghana

Regional full-immunisation coverage, district vulnerability clusters & an interpretable risk index · 16 regions / 261 districts

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Ghana's high national immunisation coverage hides a northern belt of under-vaccinated districts where structural access barriers — not demand — keep children unprotected.

75.2

Mean full coverage
% fully vaccinated (DHS 2022)

180/261

Districts below target
<80% fully vaccinated

67.5 vs 77.2

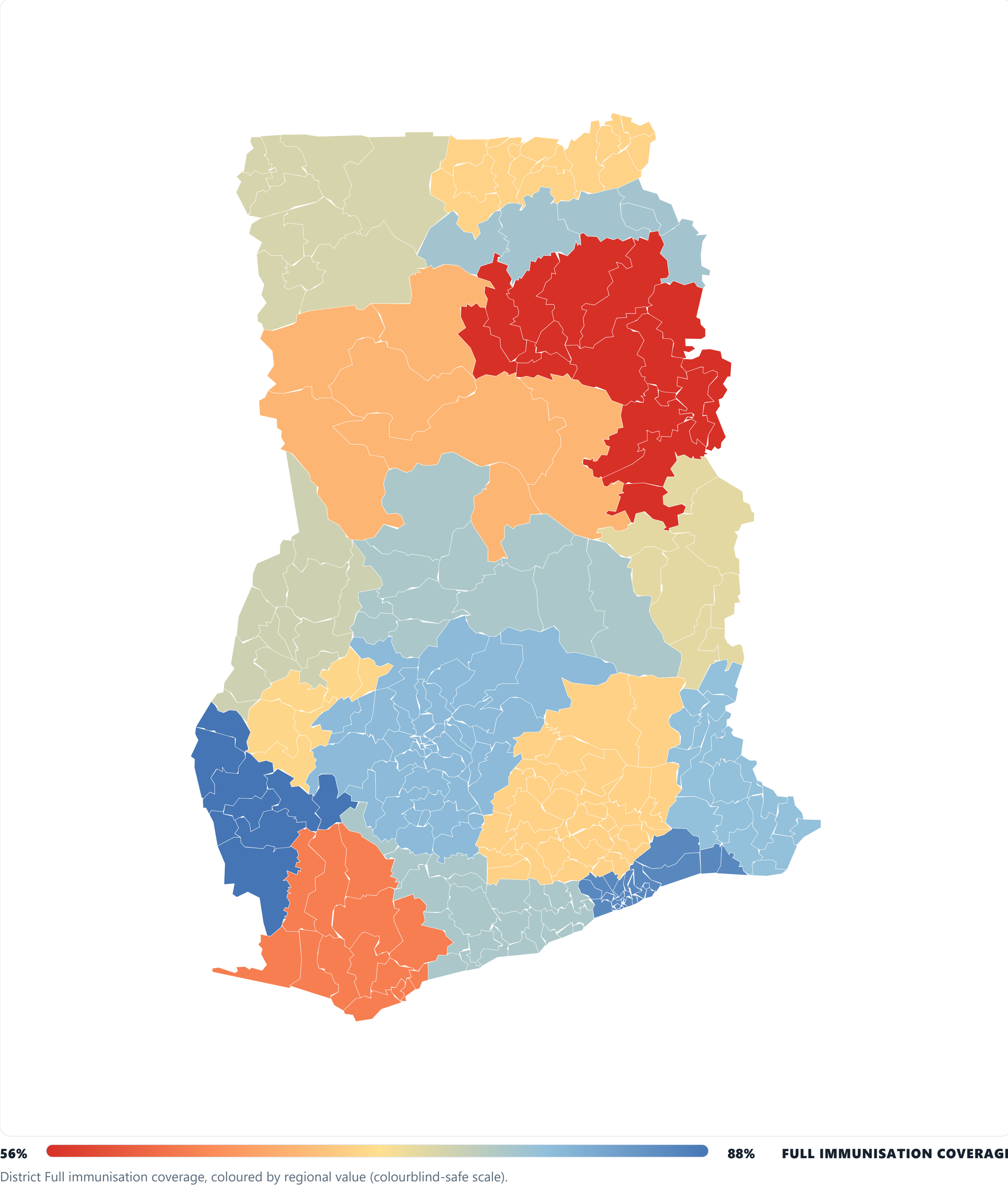
North–South coverage
Northern vs Southern belt (%)

0.72

Moran's I (IDRI)
district vulnerability clustering, $p < 0.001$

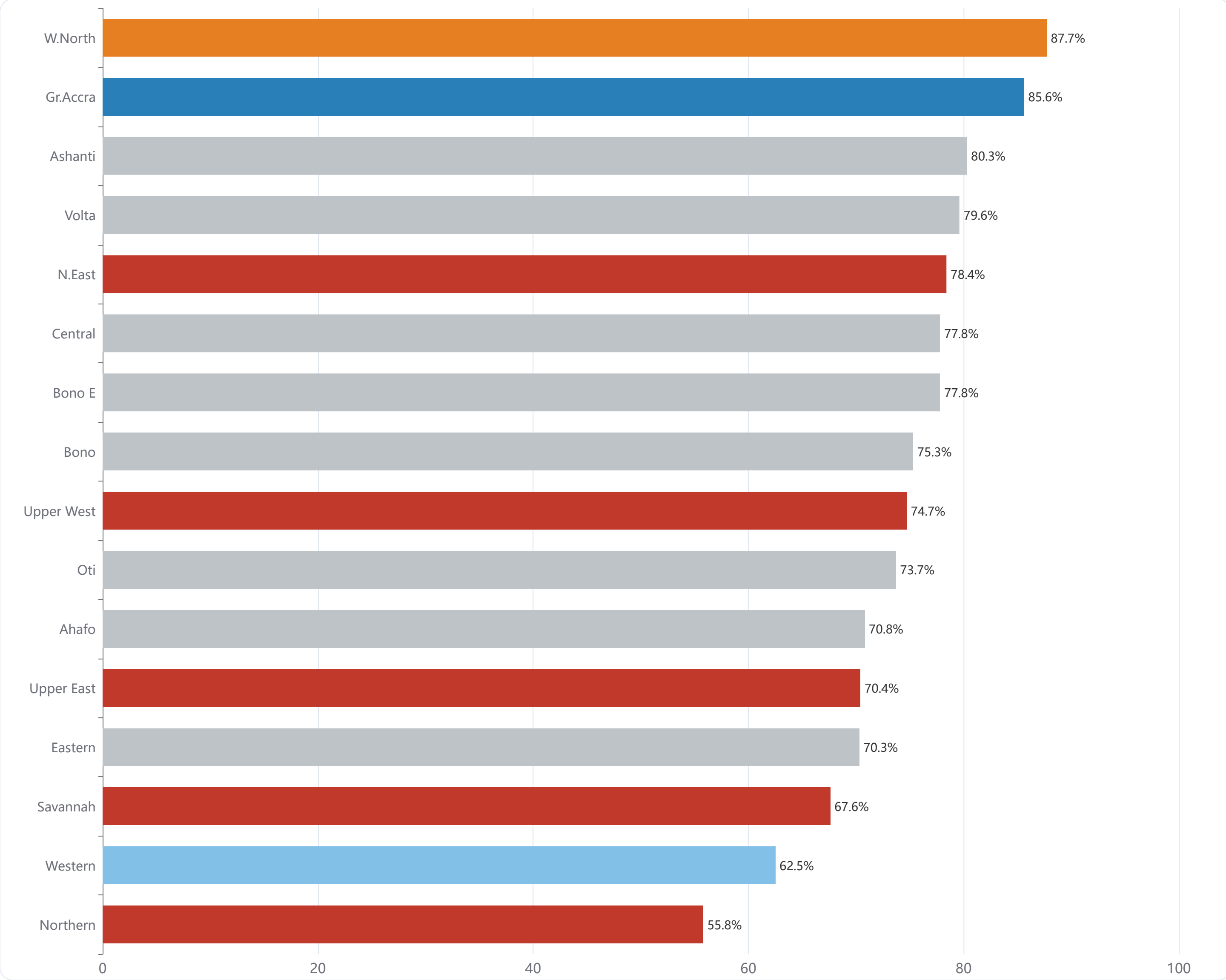
1 Where the burden concentrates

The burden is **spatially concentrated** rather than evenly spread: W.North carries the highest Full immunisation coverage (87.7%), Northern the lowest (55.8%).



2 The regional gradient

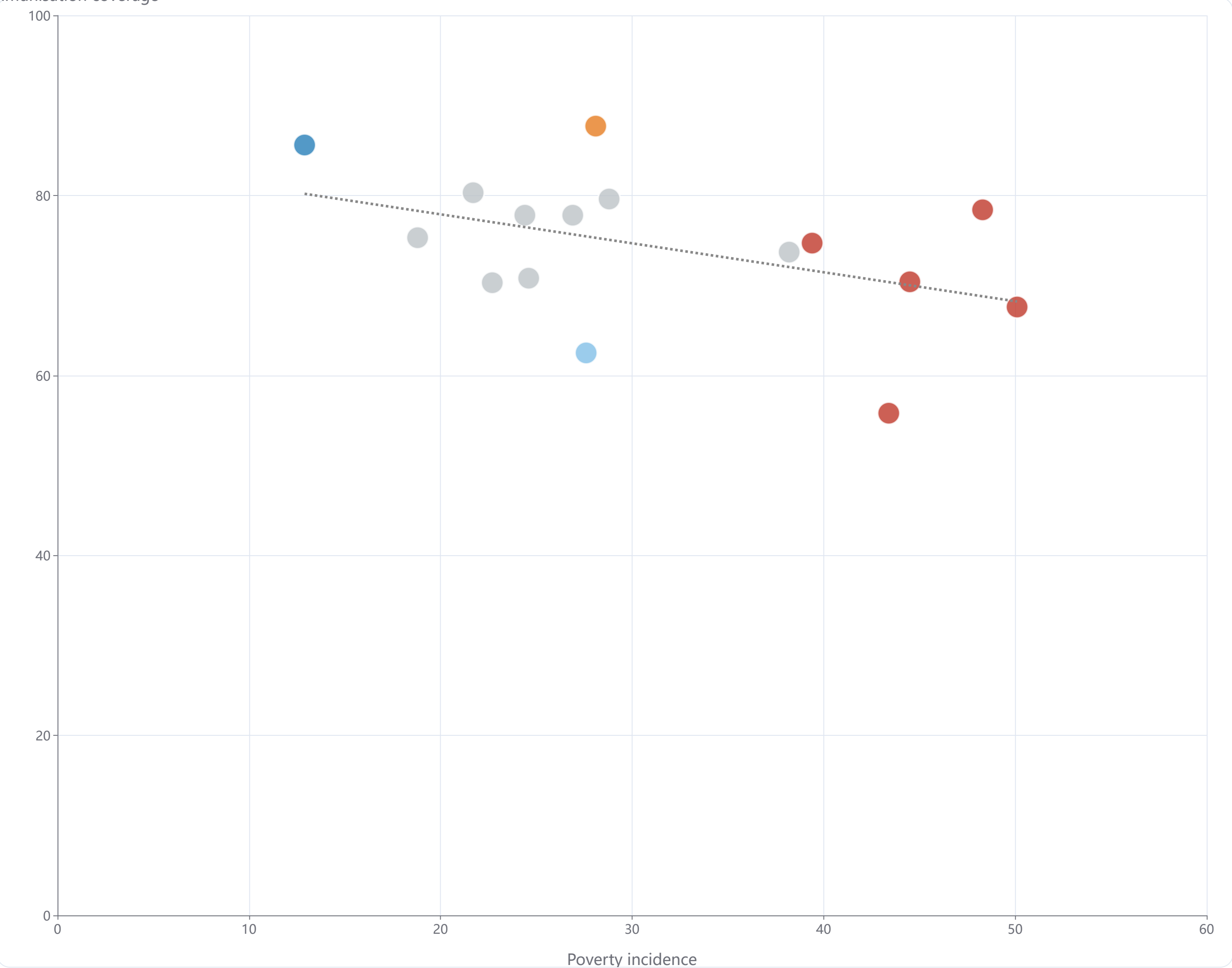
A clear **north–south gradient** separates the most- and least-affected regions, the signature of structural inequity.



Regions ranked; colour = spatial-cluster class.

3 What drives it

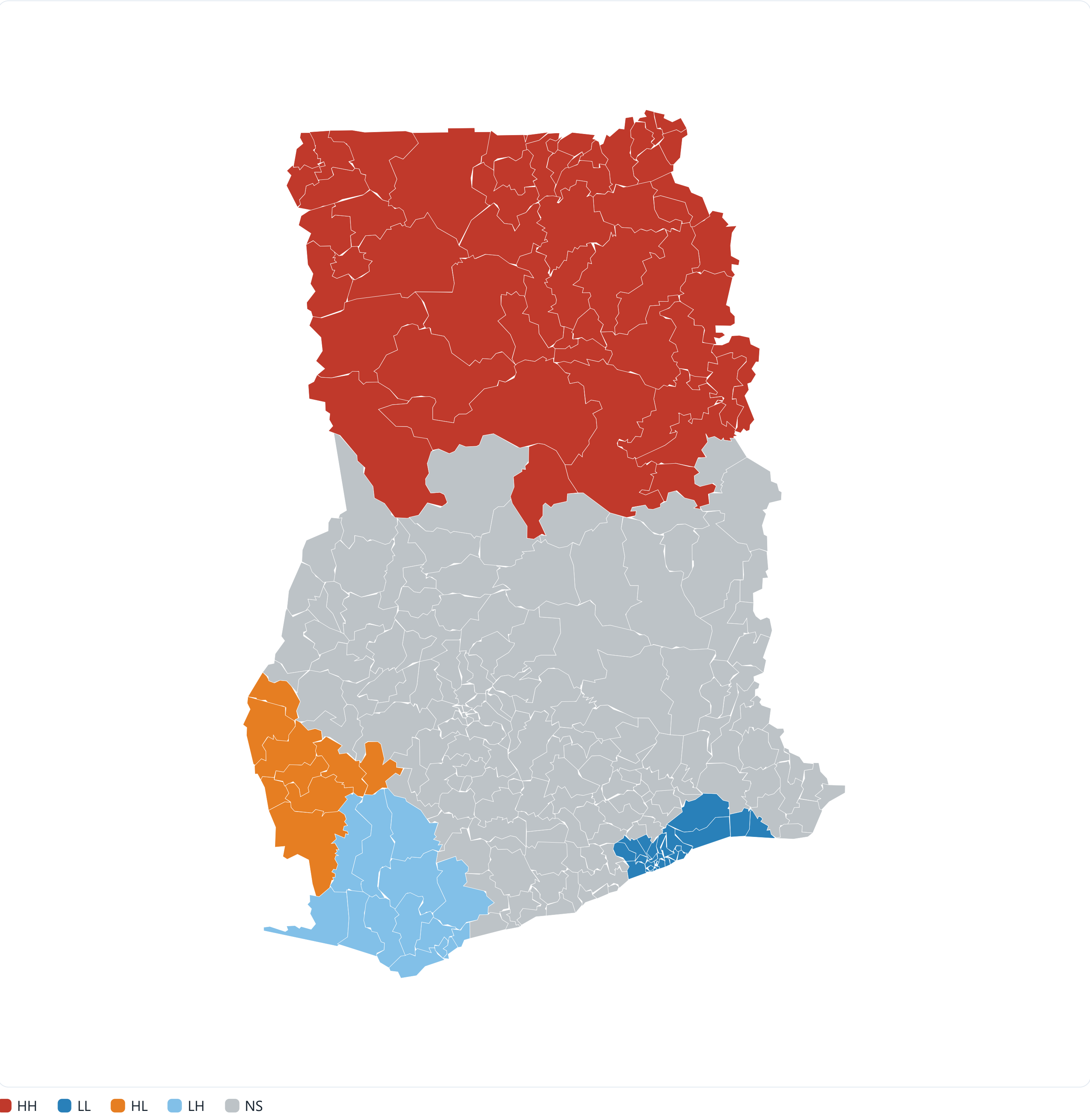
Across regions, Full immunisation coverage tracks **Poverty incidence** — the social-determinant pathway the analysis isolates.



Full immunisation coverage vs Poverty incidence, 16 regions, with OLS fit.

4 Spatial cluster structure

Local Moran's I confirms a significant **high-high cluster** and a complementary low-low cluster — the spatial fingerprint of the disparity.



What it means

Equity-weighted EPI investment in the northern high-vulnerability cluster, where >90% coverage is achievable elsewhere, is the efficient route to closing the gap.

